

Topic 2.1 - Change in Arithmetic and Geometric Sequences

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Why this topic matters

Sequences are functions whose inputs are usually whole-number indices. The strongest solutions identify the invariant that drives the sequence, state the domain, and coordinate at least two representations rather than relying on visual pattern recognition alone.

Learning targets

- Describe arithmetic change as repeated addition and geometric change as repeated multiplication.
- Move among recursive, explicit, numerical, graphical, and contextual representations.
- Use discrete domains, precise notation, and contextual units.

Language and structure

Term	Meaning in this topic
common difference	The constant amount added to obtain the next term of an arithmetic sequence.
common ratio	The constant factor used to obtain the next term of a geometric sequence.
recursive rule	A rule that defines a term from one or more earlier terms.
explicit rule	A rule that gives a term directly from its index.
discrete domain	An input set made of separated values, usually whole-number indices.

Launch: make a claim before calculating

Launch investigation

Reasoning first

Two sequences begin at 12. Sequence A adds 7 each step. Sequence B multiplies by 1.35 each step. Without calculating more than four terms, predict which sequence will eventually be larger and explain what evidence would justify the claim.

Concept 1: Arithmetic change: equal additions

Core idea

If $a_{n+1} - a_n = d$ for every allowed index, then the sequence is arithmetic.

- Recursive form: $a_1 = A$, $a_n = a_{n-1} + d$ for $n \geq 2$.
- Explicit form: $a_n = A + d(n - 1)$.
- A positive d means the terms increase by a constant amount; it does not guarantee every term is positive.

Worked example - annotate the reasoning

Problem. An arithmetic sequence satisfies $a_4 = 23$ and $a_{11} = 58$. Determine d , a_1 , and an explicit rule.

Model reasoning. The index gap is $11 - 4 = 7$, so $58 - 23 = 7d$ and $d = 5$. Then $a_1 = 23 - 3(5) = 8$. Thus $a_n = 8 + 5(n - 1) = 5n + 3$.

Check your understanding

Independent

The sequence b_n has $b_3 = -2$ and $b_9 = 34$. Find b_1 and b_n .

Concept 2: Geometric change: equal multiplications

Core idea

If $\frac{g_{n+1}}{g_n} = r$ for every allowed index with $g_n \neq 0$, then the sequence is geometric.

- Recursive form: $g_1 = G$, $g_n = r g_{n-1}$ for $n \geq 2$.
- Explicit form: $g_n = G r^{n-1}$.
- For $0 < r < 1$ and $G > 0$, the sequence stays positive and decreases toward 0.

Worked example - annotate the reasoning

Problem. A positive geometric sequence has $g_2 = 12$ and $g_5 = 324$. Determine r , g_1 , and g_n .

Model reasoning. The index gap is 3, so $324/12 = r^3 = 27$ and $r = 3$. Then $g_1 = 12/3 = 4$, so $g_n = 4(3)^{n-1}$.

Check your understanding

Independent

A geometric sequence has $c_2 = -6$ and $c_5 = 48$. Find the real value of r and an explicit rule.

Concept 3: Representations and domain

Core idea

A finite list supports a model only after a sequence family or defining rule is stated.

- Sequence graphs use isolated points because intermediate indices are not automatically allowed.
- A table can reveal constant differences or ratios, but several formulas can agree on finitely many displayed terms.
- In context, the index, initial value, and units determine whether the first term is a_0 or a_1 .

Worked example - annotate the reasoning

Problem. A savings account begins with 500 QAR at month 0 and grows by 8% each month. Write recursive and explicit rules, then state the natural domain for the first year.

Model reasoning. $A_0 = 500$ and $A_n = 1.08A_{n-1}$. Explicitly, $A_n = 500(1.08)^n$. For the first year the natural domain is $n \in \{0, 1, \dots, 12\}$, measured in whole months.

Check your understanding

Independent

Explain why connecting the points of the sequence $s_n = 2^n$ can create a misleading graph.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Classify $18, 12, 8, \frac{16}{3}, \dots$ and write an explicit rule beginning at $n = 1$.

Exit ticket 2

No calculator

An arithmetic sequence has $u_7 = 41$ and common difference -4 . Find u_1 .

Exit ticket 3

No calculator

State one reason the terms $2, 5, 10, 17$ do not by themselves prove a unique continuation.
